

designs inspired by the fins of whales and a lot more. Sometimes, these are serendipitous. Being an observant lot, scientists and engineers notice a lot of interesting things around them. Some of these fascinating sights might suddenly strike a loud gong in their heads and lead to a great invention. But, it is not always a stroke of luck!

"Since we are part and parcel of nature, inspiration from nature has been used for ages. There is nothing new there. My own work leading to sculptured thin films, which are nano-engineered meta-materials, was inspired by a specimen of *ulixite*, a mineral sometimes sold as 'TV rock'. Bio-inspiration is usually serendipitous. Biomimetics, on the other hand, can be either forced or serendipitous. For instance, the invention of Velcro was serendipitous, but robotic mountain climbers designed to mimic the gait of mountain goats and miniature flyers designed to emulate hummingbirds are examples of deliberately seeking solutions from nature to fulfil specific requirements," explains Dr Akhlesh Lakhtakia, the Charles G. Binder (Endowed) Professor of Engineering Science and Mechanics, Pennsylvania State University.

In the latter type of cases, once it is decided that certain natural solutions have to be emulated, researchers follow the usual patterns. They observe the natural solution, consult with colleagues in relevant biological sub-disciplines, isolate essential characteristics and mechanisms of the natural solution and then work out emulation strategies, keeping in mind that

Commercial successes

Dr Judith Braganca of BITS cites some examples of nature-inspired technologies that have gone on to become commercially-successful products:

1. AutoFlex MARAG, based on the structural surface features of moth eyes, is an optical thin-film coating that eliminates light reflection and glare. Applications include coatings for PDAs and solar cells. The 20µ optically perfect GRIN lenses found on the exoskeleton of certain brittle sea stars and the glass fibres found on a particular glass sea sponge, are influencing the development of micro-lenses and fibre optics respectively.
2. Ultracane, inspired by the bat's use of ultrasound, provides a blind person with a more comprehensive coverage of the spatial environment than afforded by the traditional cane.
3. The Libelle High-G flight suit, inspired by the fluid-control mechanics of dragonflies, has extended the ability of pilots to maintain function while experiencing high G forces.
4. LogiTek's Marble trackball, a wireless computer mouse, was inspired by the fly's image processing system.
5. BioAvert, an object-avoidance software program, is based on the neural network behaviour of cockroaches.
6. Super Microft, an award winning fabric developed by the Teijin Group of Japan, was developed by mimicking the water-repelling properties of the lotus leaf including the surface micro-texture.

a natural solution is not necessarily more efficient or economical than an artificial one. The scrutiny of natural phenomena, organisms, structures and such to solve problems, create better designs or develop new products, is developing into an organised field of study. In fact, universities such as the MIT have included biomimetics as part of their curriculum.

Delving upon whether bio-inspiration is by chance or by design, we can outline several approaches to nature-inspired technology development.

Approach 1

"This is interesting; let me study it in detail."

Picking up a stray clue and turning it into a discovery or invention is one of the most admirable things to do. An apple fell on Newton's head and he discovered gravity, we say. However, Newton would admit it was not that easy! He discovered gravity because he questioned why the apple fell, and pursued to study the reasons behind it.

A researcher sees something interesting, is inspired by it, studies it further, and tries to apply that knowledge

to develop a theory, design or produce or whatever other fragment of technology. That is, the researcher is not in search of an idea, but chances upon one, and decides to pursue it.

The sculptured thin film mentioned by Dr Lakhtakia is one such nature-inspired technology. About two decades ago, he observed a mineral called *ulixite* which has a fibrous structure. It is often called TV rock because of its ability to transmit images as on a television screen, when cut and polished. Intrigued by this, Dr Lakhtakia asked a colleague whether such fibrous structures could be made with thin-film technology, and was glad when told that they could be!

He decided to lay down the fibres of the thin film at an oblique angle, to try and find out the mathematical basis of its dielectric, magnetic and magneto-electric properties. Then, he decided to shape the fibres so that they were not straight. He found that the chirality, or left- or right-handedness of the film, affects the transmission of light. If the handedness of the light and the film are same, it can be very bright. If they are opposite, the transmission is low. He also showed that it was possible to shift the colour of the film after deposi-



LogiTek Marble trackball—a wireless computer mouse inspired by the fly's image processing system